

MedTrack_DV - Dashboard Storyboard & Wireframe Specification

Author: Healthcare Data Analytics Team | **Deliverable:** Module 4 Prototype Specification

1. Executive Summary & Workbook Architecture

The **MedTrack_DV** Tableau Workbook is designed as an interactive, executive-ready dashboard suite for hospital administrators, clinical operations managers, and department heads. The workbook unifies four core operational views into a single, cohesive user interface connected via global filters and parameter actions.

Dashboard Name	Core Focus & Key Performance Indicators	Target Audience
Hospital Overview	Total Admissions, Occupancy Rate %, ALOS, Readmission %, Bed Utilization %	C-suite, Medical Directors
Patient Flow	Admission vs. Discharge trends, Stay Duration distribution, Peak Load tracking	Operations Managers, Triage
Department Analytics	Department Volume, Efficiency Scores, Cost per Patient, Readmission by Specialty	Department Heads, Chief of Surgery
Resource Utilization	Bed Utilization %, Staff Allocation Ratio, Equipment Usage (MRI, Ventilators)	Resource Allocation, Nursing Lead

2. Global Interactivity, Navigation & Parameter Actions

Global Filter Bar: Displayed on top of every dashboard container for synchronized cross-filtering:

- Date Range Filter:** Slider & Dropdown (Default: Full Year 2024).
- Hospital / Region Filter:** Multi-select dropdown (City Care, Green Valley, Sunrise, Metro Health, HealthPlus).
- Department Filter:** Multi-select dropdown (Cardiology, Surgery, Pediatrics, ICU, Emergency, Gen Med, Orthopedics).
- Patient Type Filter:** Inpatient, Outpatient, Emergency, Day Care.

Navigation Actions: Embedded top tab bar with active state highlights allowing seamless switching between Overview, Patient Flow, Department Analytics, and Resource Utilization.

3. Dashboard 1 Wireframe: Hospital Overview

Layout Grid (1920 x 1080 Desktop Standard):

Top KPI Bar (6 Cards)	Total Admissions (5,015) Occupancy Rate (82.1%) ALOS (5.1 Days) Readmission Rate (13.0%) Bed U
Main Chart 1 (Left 60%)	Monthly Admission & Discharge Trends (Dual Line Chart with YoY Comparison)
Main Chart 2 (Right 40%)	Occupancy & Bed Utilization Trend (Area Chart with Threshold Indicator @ 85%)
Bottom Left (50%)	Readmission Rate by Specialty & Severity (Bar Chart)
Bottom Right (50%)	Regional Hospital Performance Comparison (Geographic Map / Ranked Horizontal Bars)

4. Dashboard 2 Wireframe: Patient Flow Analytics

Layout Grid & Visualizations:

Top Summary	Patient Volume Flow Indicator Average Stay Metric Cards Peak Load Alert Banner
Upper Container	Monthly Admissions vs. Discharges Flow (Stacked Area Chart showing Net Inflow/Outflow)
Middle Container	Length of Stay (LOS) Distribution by Severity & Department (Box Plot / Histogram)
Lower Container	Peak Patient Admission Load Heatmap (Days of Week vs. Hours/Months)

5. Dashboard 3 Wireframe: Department Analytics

Top Parameter Selector	Select Primary Department vs. Benchmark Department (Dynamic Parameter Action)
Left Matrix	Department Efficiency Score vs. Patient Satisfaction Scatter Plot
Right Breakdown	Treatment Cost per Patient & Avg Length of Stay Bar Charts
Bottom Matrix	Readmission Rate & Severity Breakdown Grid per Specialty

6. Dashboard 4 Wireframe: Resource Utilization

Top KPI Gauge	Bed Utilization Rate Gauge (82.1%) Staff Ratio Gauge Critical Equipment Active %
Upper View	Bed Capacity & Occupancy Heatmap by Hospital Wing / Department
Lower Left	Staff Allocation Monitoring (Doctor/Nurse-to-Patient Ratio Bar Chart)
Lower Right	High-Value Equipment Usage (Ventilator, MRI, CT Scanner Distribution Donut Chart)